5th **BATTERY GIGAFACTORY** Summit USA

Advances in Planning, Engineering, and Operations

Phoenix, Arizona | March 11-12, 2026

AI Orchestrated Battery Manufacturing: from 5MIE to MMM

Daniel Li
Battery Manufacturing Executive

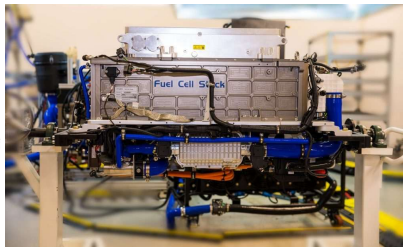
EV & Battery Manufacturing Experiences

2007-2021: 14 years in Wanxiang America Corp to build Wanxiang Group's global clean energy platform

- ✓ Executive VP of Wanxiang Clean Energy USA
- ✓ President of A123 Systems
- ✓ Board of Directors – Karma Automotive

2021-2023: Hyzon Motors

- ✓ Senior VP of Power Solutions
- ✓ President of Hyzon China



2023-2025: SES AI

- ✓ Chief Manufacturing Officer: AI for Manufacturing

2026-: ManufAgent Solutions

- ✓ Managing Director

ManufAgent™

Manufacturing Excellence: 5M1E Framework

5M1E Lens

- Man
- Machine
- Material
- Method
- Measurement
- Environment

Building & Operating a Gigafactory

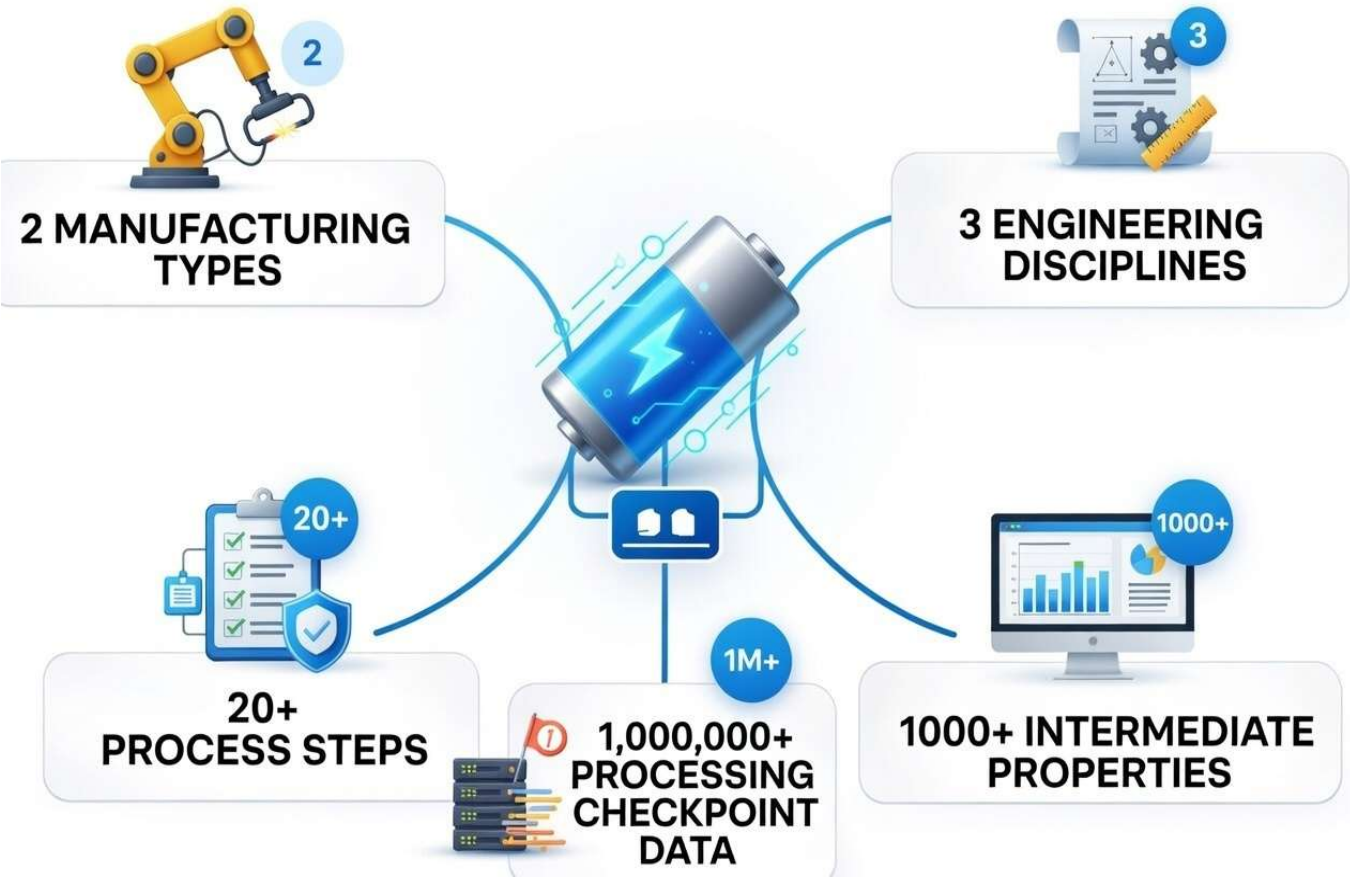
Success Depends On

- Man – Skilled and safety-compliant workforce
- Machine – Reliable, adaptive equipment
- Material – Stable and traceable supply
- Method – Standardized digital processes
- Measurement – Real-time data visibility
- Environment – Controlled, stable conditions

Gigafactory Challenge

Breakdown in any pillar drives waste, delays, or drift

Complex Gigafactory Operations



Towards Gigascale Maturity

Pattern

Misalignment

+

Immaturity

+

Data Fragmentation

=

Scrap & Delay

Typical early gigafactory yield: 60–80%

Success Formula

Align all 5M1E pillars

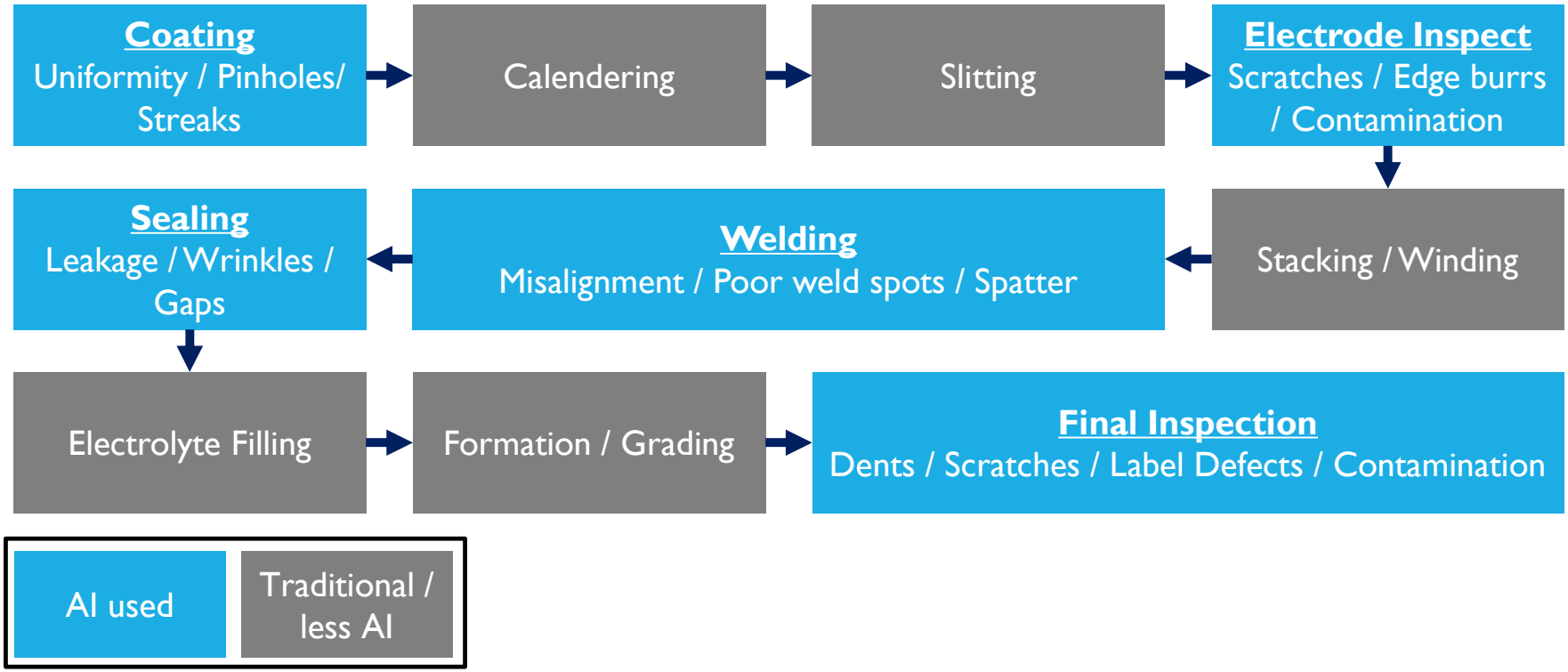
+

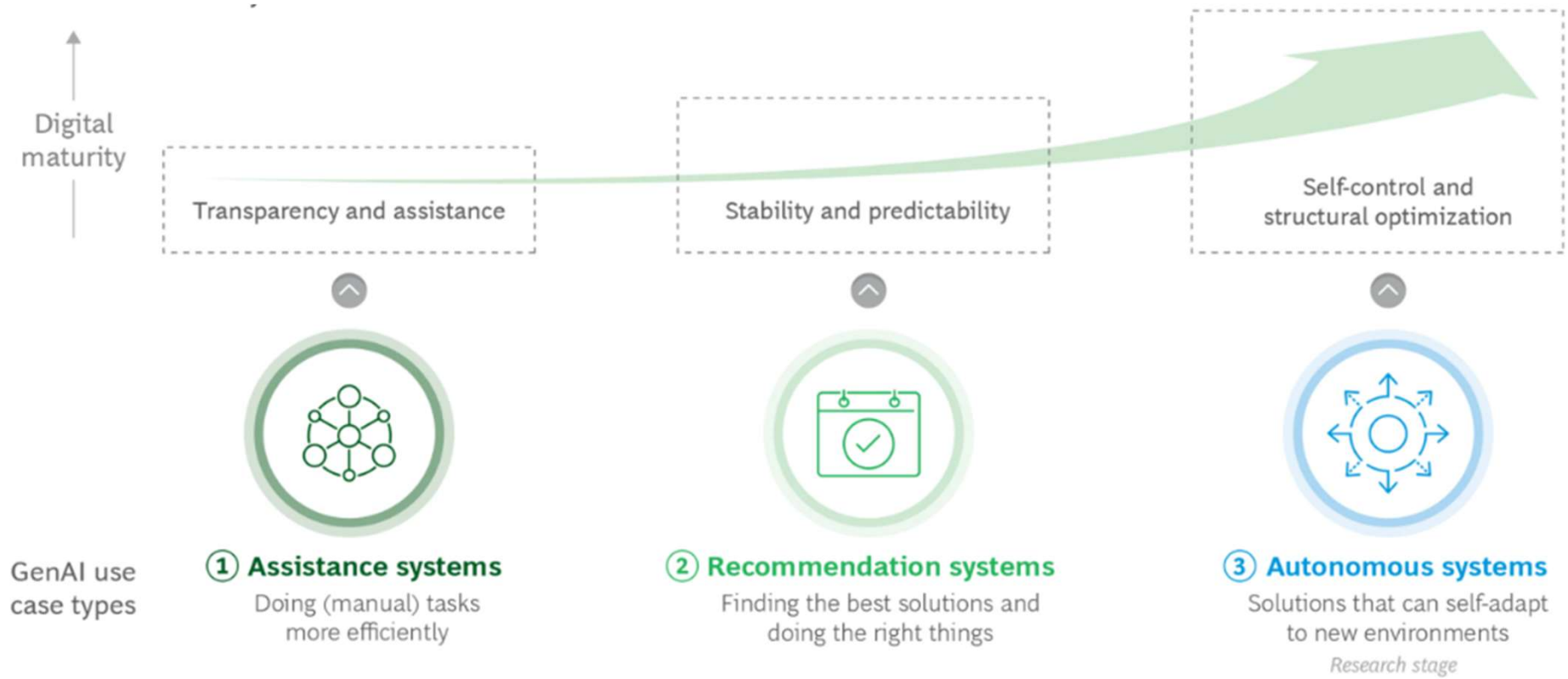
Standardize data and interfaces

+

Evolve into adaptive, digital-first manufacturing systems

AI in Battery Manufacturing





GenAI Supports the Factory of the Future

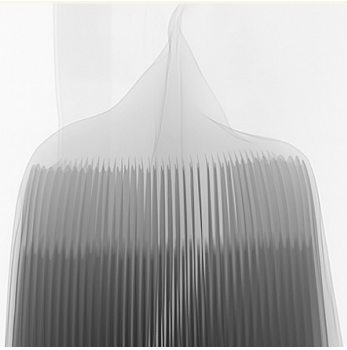
Generative AI's Role in the Factory of the Future | BCG

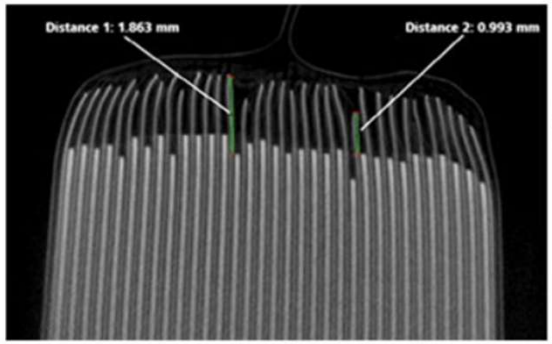
AI Transforms the 5M1E Framework into a Self-Learning System

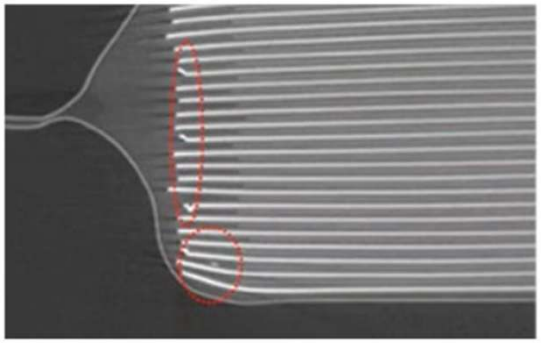
AI connects the 5M1E from data to decision

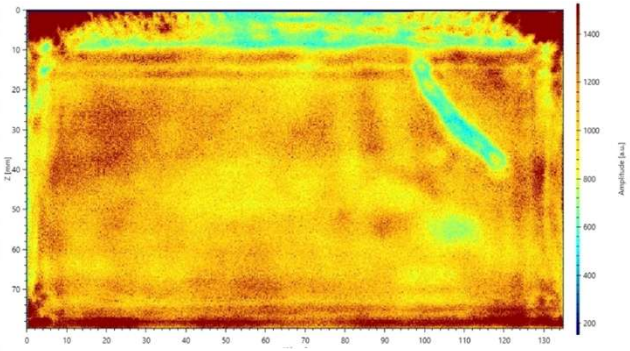
GenAI translates intelligence into human-centered action

5M1E	AI Transformation	Result
Man	GenAI copilots, digital SOPs, human–machine collaboration	Empowered, data-driven workforce
Machine	Predictive maintenance, adaptive control	Higher uptime, optimized yield
Material	Traceability via AI vision, predictive supply chain	Lower waste, stable flow
Method	Digital twins, AI optimization, GenAI process design	Continuous improvement, agility
Measurement	Intelligent vision, data fusion, AI root cause analysis	Faster feedback, zero defect quality
Environment	Energy optimization, emission tracking, AI simulation	Sustainable, controlled operations









X-Ray

Detection of A/C overhang

Image credit: Comet X-ray

Industrial CT

High-resolution image for clear detection of overhang
Detection of bent electrodes and a metal particle

Image credit: ZEISS

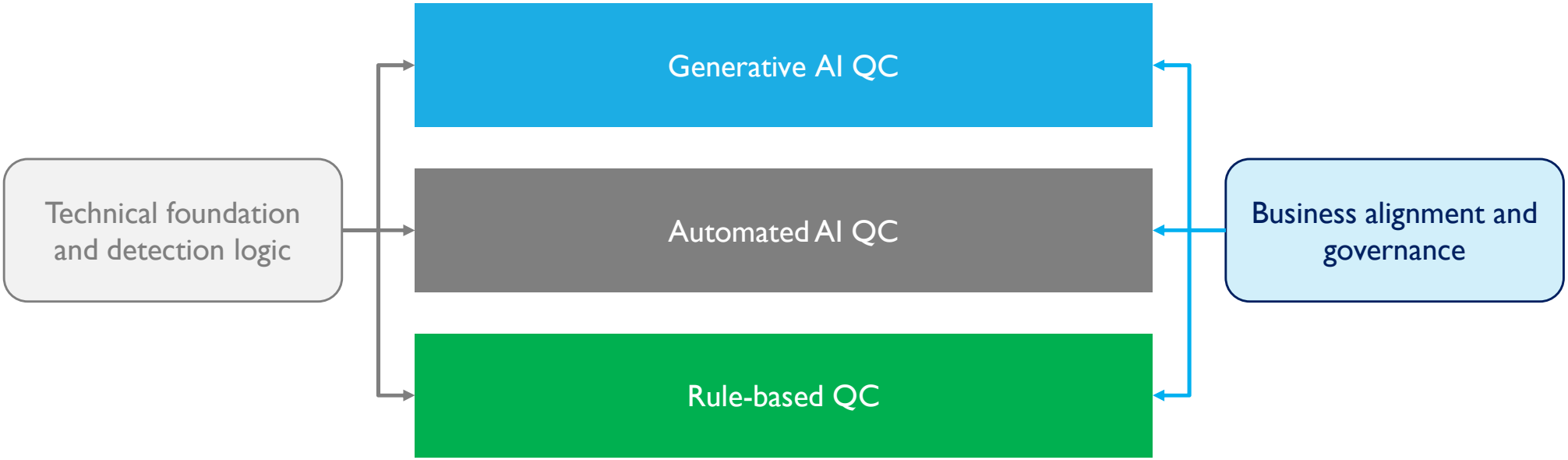
Ultrasound

Inspection of Wetted, Formed, and Aged Cells
Cell's electrolyte distribution & gas presence

Image credit: XARION

Case Highlights: AI EoL Detection

Predictive defect detection model, trained on full set of cell data to catch flawed cells.
Explainable, root cause analysis. Touch-free, no additional capex for QC devices



Capabilities progress bottom-up: from rules → automation → generative intelligence

Quality Control + AI: near 0 defects

Quality is not just a manufacturing KPI — it is a **strategic business risk**

Battery Defects Cause Severe Consequences

Typical EV recall cost:
- $500M – $2B depending on scale

CAR AND DRIVER

e.g. 2025 Toyota RAV4

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VW Is Recalling Nearly 44K ID.4s Due to Batteries That May Catch Fire

The recall involves 2023 to 2025 models with defective battery modules, which can be fixed with a software update.

By Eric Stafford Published: Feb 2, 2026

Reuters World

Exclusive: Volvo Cars to recall 40,000 electric SUVs over battery fire risk

By Marie Mannes

February 23, 2026 9:34 AM CST · Updated February 23, 2026

Aa

Replacement battery modules could cost $195MM

Evolution

AI-Powered Agile Lean Manufacturing

Yesterday:

Efficiency and stability

Market trend:

Customized

High Mix Low Volume

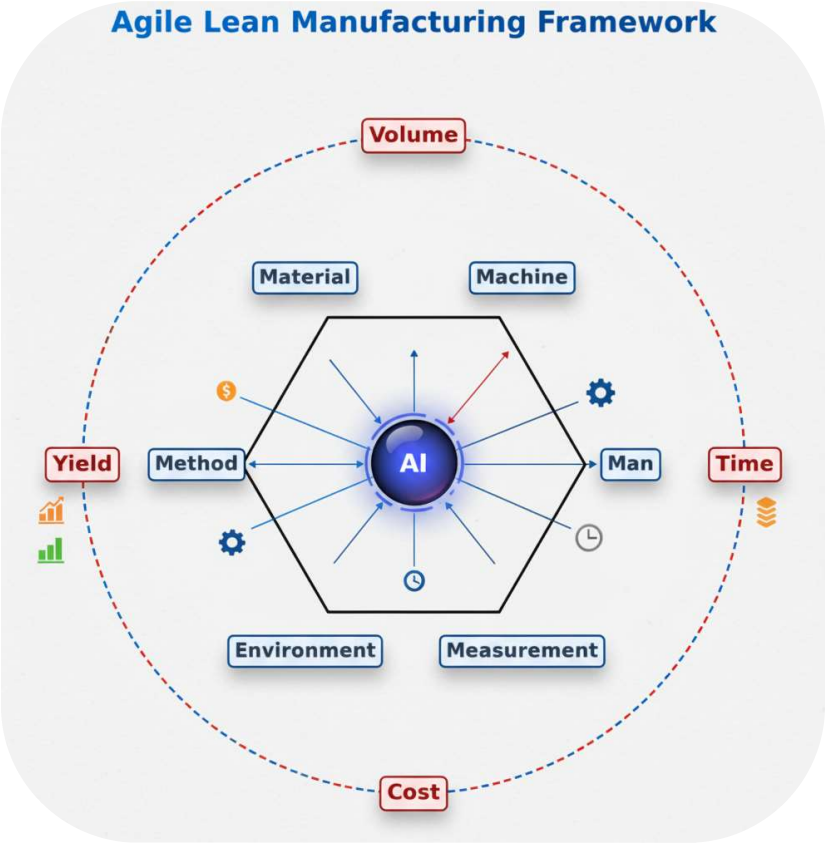
Today:

Agility and intelligence

AI powers real-time adaptation

4 Measurable KPIs: TVYC

Time, Volume, Yield, Cost

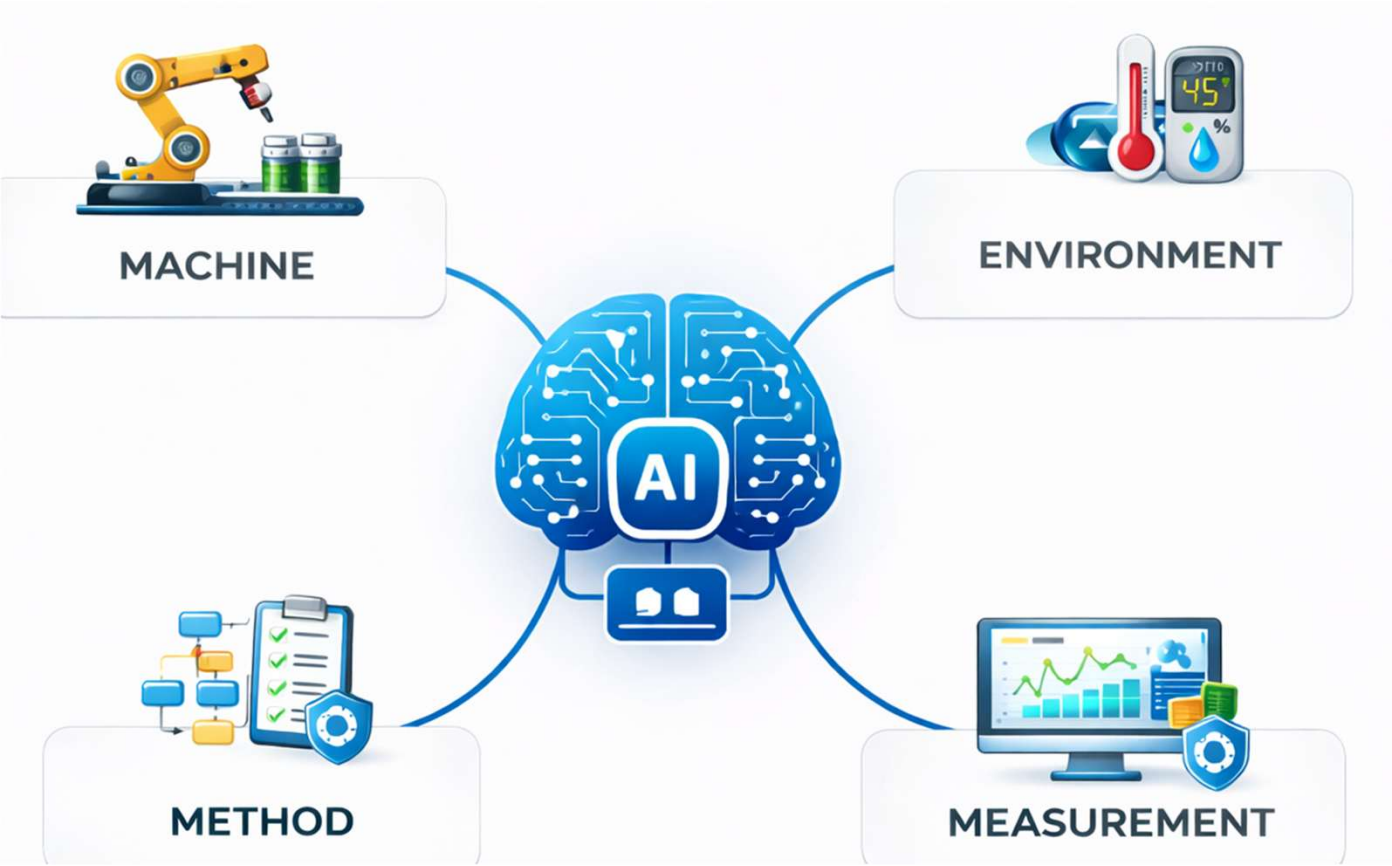


AI-Orchestrated Manufacturing

Complexity → Simplicity

AI converts factory complexity into software-defined operations:

***Four** pillars 3M1E become **One** system*



Traditional Factory

5M1E:
Man
Machine
Material
Method
Measurement
Environment

Agentic AI

ManufAgent
integrates
Machine, Method,
Measurement, and
Environment

AI Factory

5M1E → MMM:
Man
Material
ManufAgent

AI Transforms 5M1E → MMM

Human + AI Collaboration = Agile Lean Manufacturing

Humans innovate/govern,

AI orchestrates/executes

Together they adapt, learn, and evolve

ManufAgent amplifies human capability

🏭 **ManufAgent**

=

🚗 **Autonomous Car**

ManufAgent *is to manufacturing what the autonomous car is to mobility*

Gigafactories that can learn to excel

Gigafactories are no longer mechanical systems — they are cyber-physical systems.

Thank You
— FOR LISTENING —
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Outtinesion
Atarfelont
MANUFAGENT
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MANUFA